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# Radxa ROCK 2A

schematic

V1.20

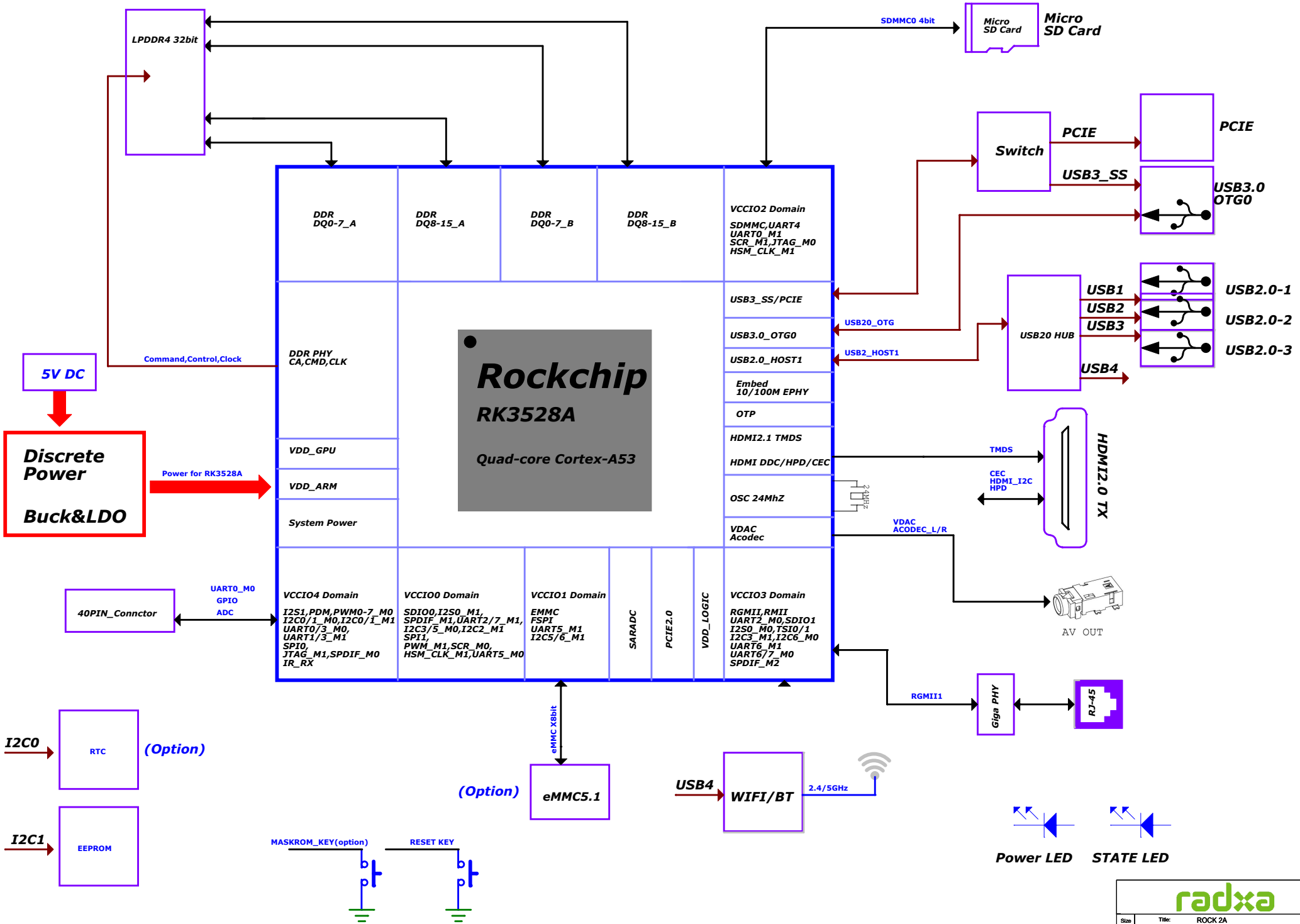
# Index


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| Page 24 | 24.5V_Power |
| Page 25 | 25.Key_LED_RTC_EEPROM |
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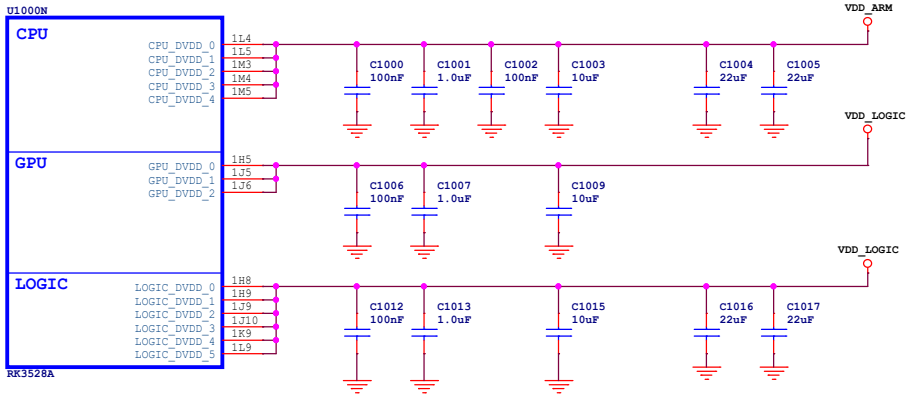
| Size | Title: | REV |
|---|---|---|
| A4 | ROCK 2A | V1.2 |
| | Page Name: 02.Index | |
| Date: Thursday, June 06, 2024 | Sheet 02 of 25 | |

# RK3528A Block Diagram

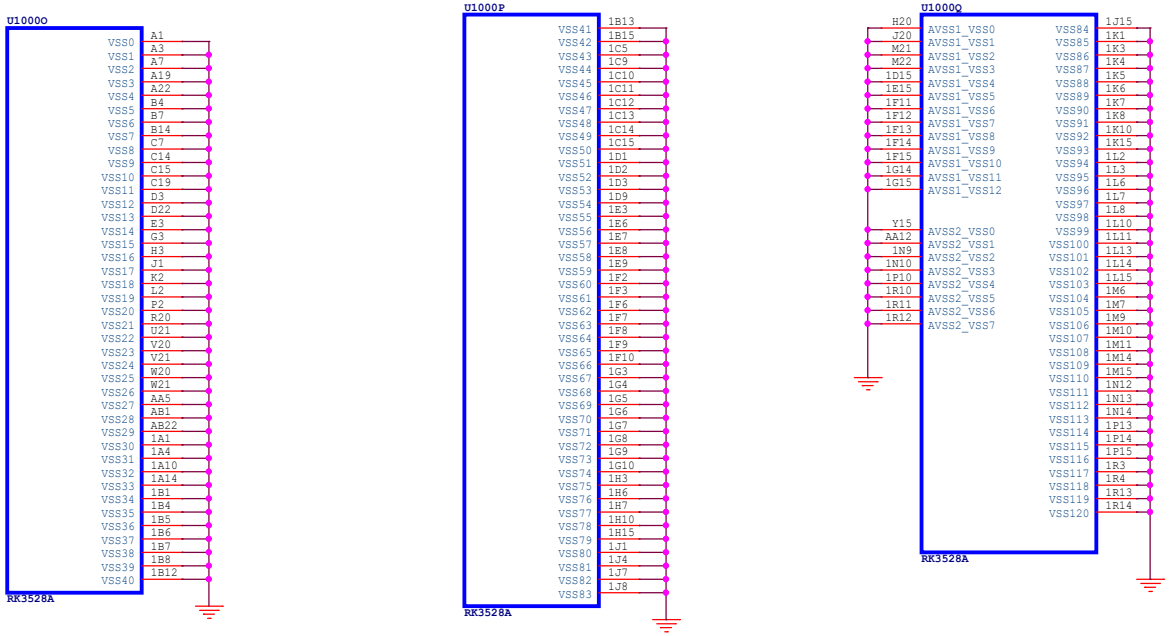



| | | |
|---|---|---|
| Size | Title: ROCK 2A | REV |
| A2 | Page Name: 03.Block | V1.2 |
| Date: Thursday, June 06, 2024 | Sheet 03 of 25 | |

# RK3528A_N(POWER)

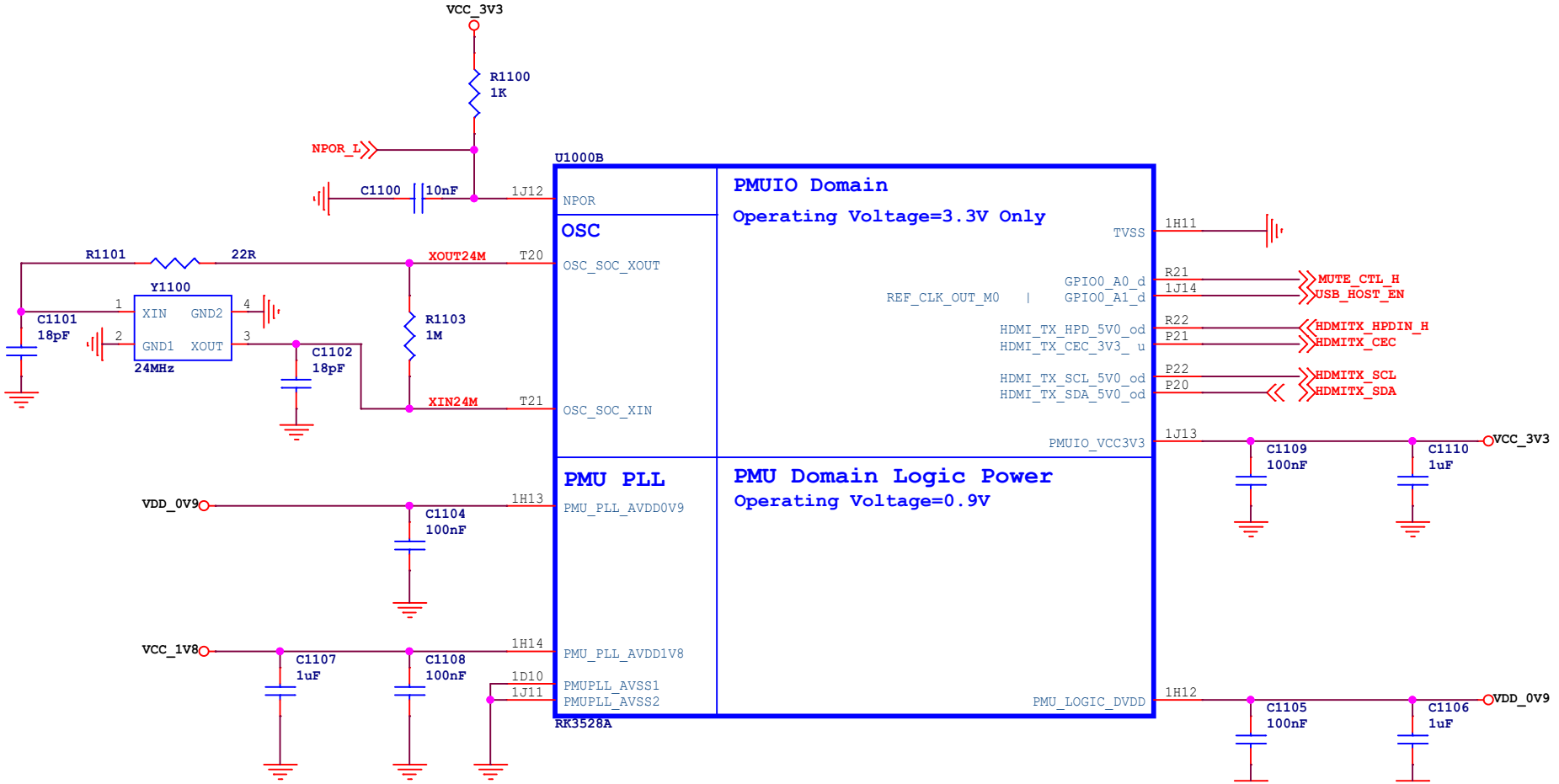



# RK3528A_O/P/Q(GND)



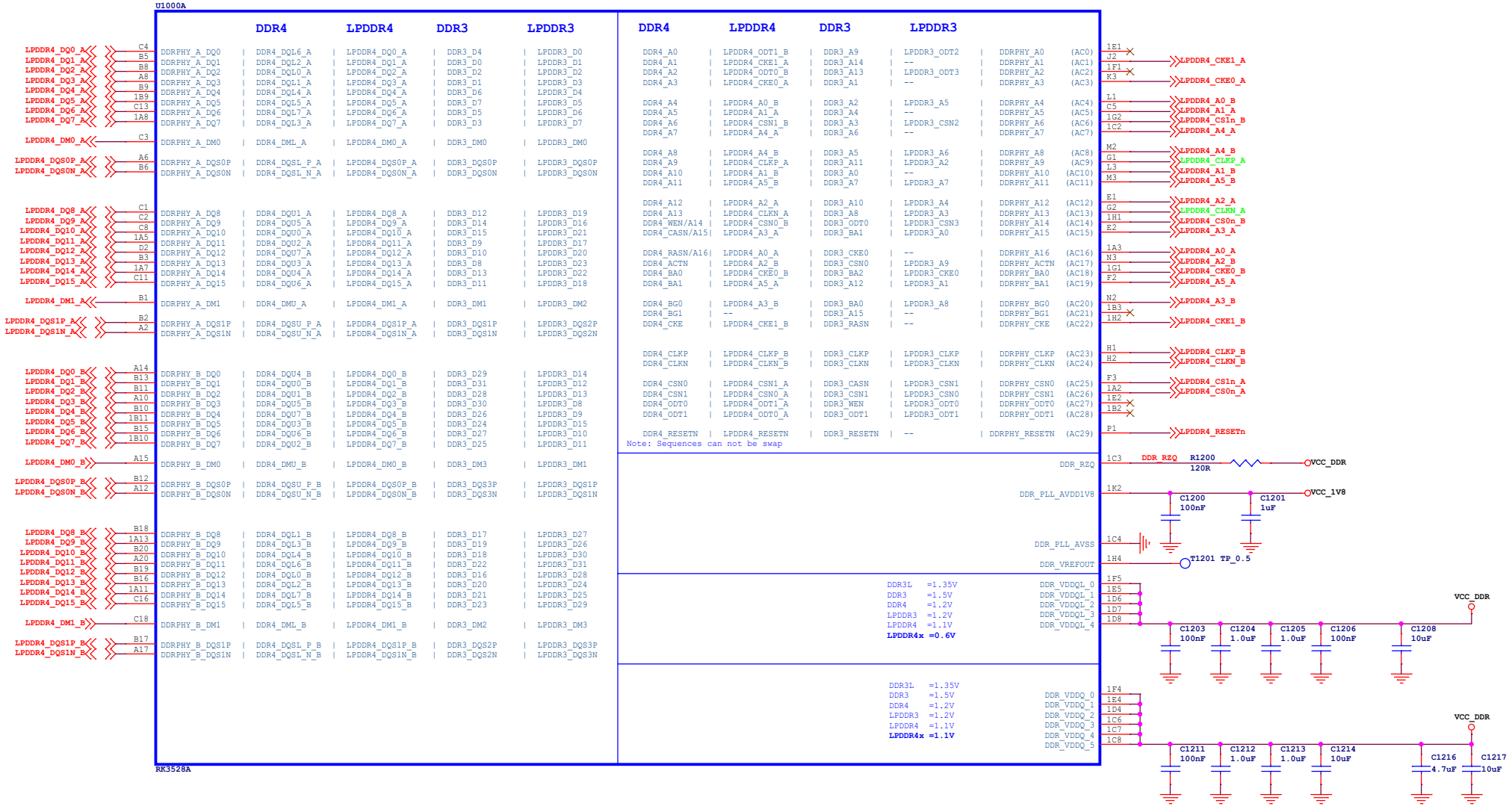

| | | |
|---|---|---|
| Size A3 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 04.RK3528A Power/GND | |
| Date: Thursday, June 06, 2024 | Sheet 04 of 25 | |

# RK3528A_B(OSC/PLL/PMUIO Domain)



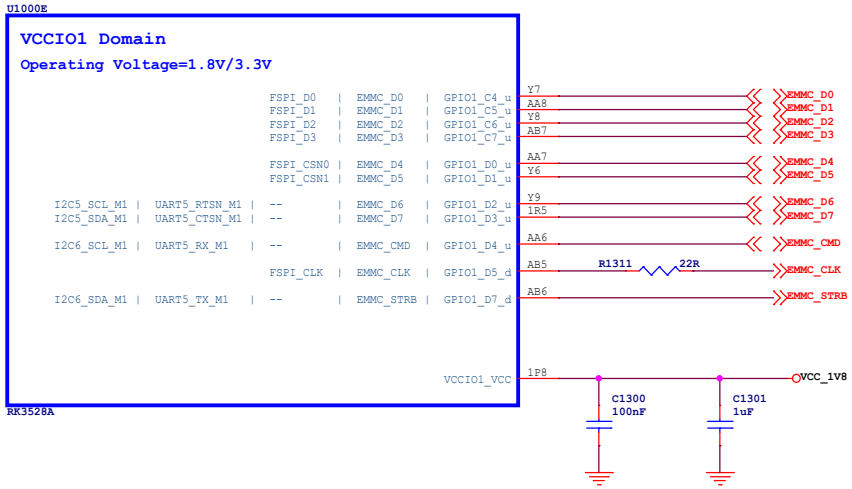

| | | |
|---|---|---|
| Size A4 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 05.RK3528A OSC/PLL/PMUIO | |
| Date: Thursday, June 06, 2024 | Sheet 05 of 25 | |

# RK3528A_A(DDR PHY)

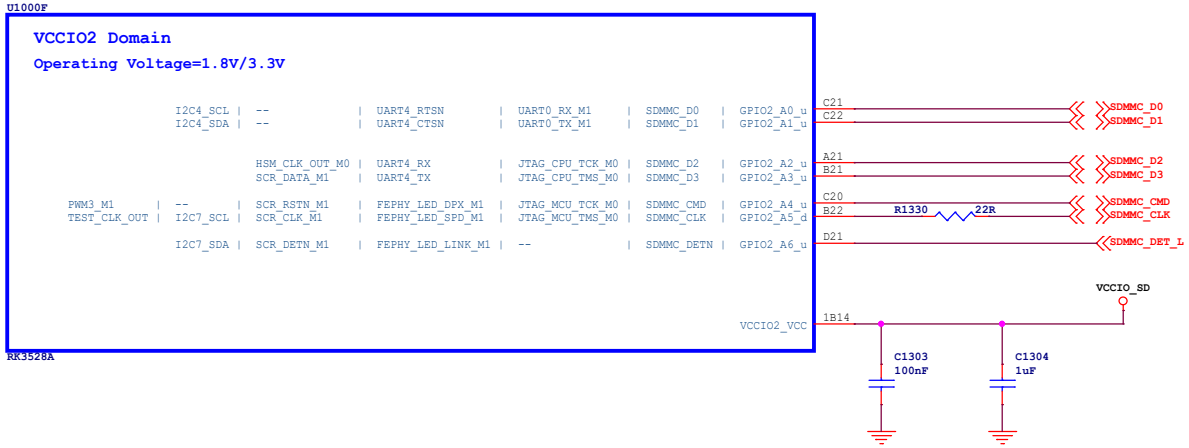



| | |
|---|---|
| Size | A3 |
| Title | ROCK 2A |
| Page Name | 06.RK3528A DDR Controller |
| REV | V1.2 |
| Date | Thursday, June 06, 2024 |
| Sheet | 06 of 25 |

# RK3528A_E(VCCIO1 Domain)

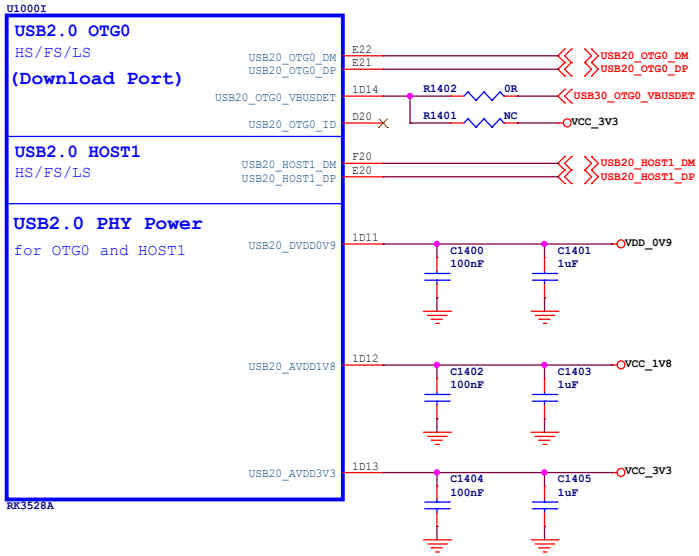



# RK3528A_F(VCCIO2 Domain)

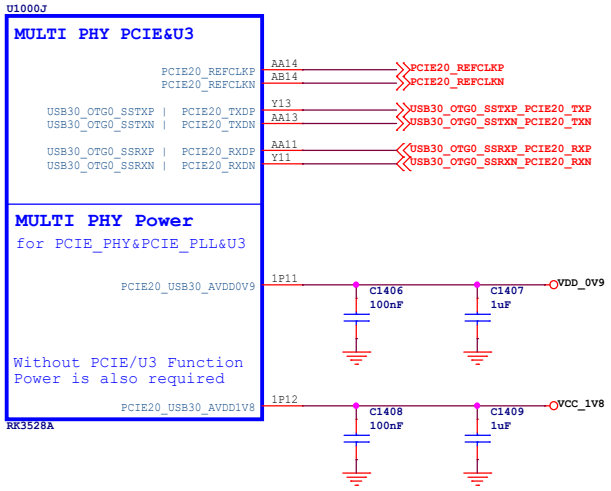



| | | |
|---|---|---|
| Size: A3 | Title: ROCK 2A | REV: V1.2 |
| | Page Name: 07.RK3528A FLASH/SD Controller | |
| Date: Thursday, June 06, 2024 | Sheet 07 of 25 | |

# RK3528A_I(USB2.0 OTG/HOST)



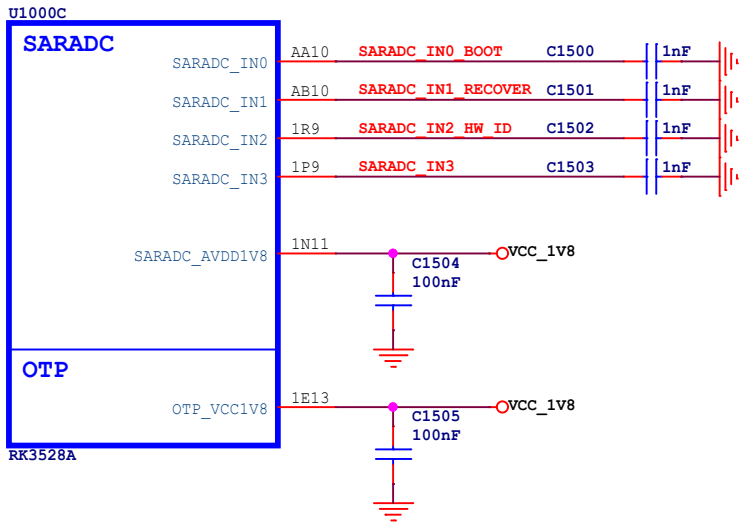

# RK3528A_J(PCIE2.0/U3 PHY)




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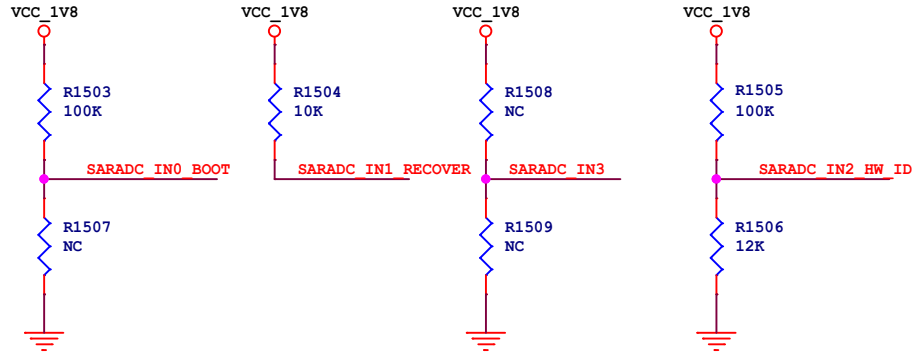
| Size | Title: | ROCK 2A | REV |
|---|---|---|---|
| A3 | Page Name: | 08.RK3528A USB/PCIE Controller | V1.2 |
| Date: | Thursday, June 06, 2024 | Sheet 08 of 25 | |

# RK3528A_C(Saradc/OTP)




**TABLE**

| Item | Rup | Rdown | ADC | BOOT MODE |
|---|---|---|---|---|
| LEVEL1 | DNP | 100K | 0 | USB (Maskrom mode) |
| LEVEL2 | 100K | 12K | 114 | |
| LEVEL3 | 100K | 27K | 228 | FSPI--USB |
| LEVEL4 | 100K | 51K | 342 | |
| LEVEL5 | 100K | 82K | 456 | |
| LEVEL6 | 100K | 120K | 570 | EMMC--USB |
| LEVEL7 | 100K | 200K | 683 | EMMC--SD Card--USB |
| LEVEL8 | 100K | 330K | 796 | SD Card--USB |
| LEVEL9 | 100K | 820K | 910 | |
| LEVEL10 | 100K | DNP | 1023 | FSPI--EMMC--SD Card--USB |



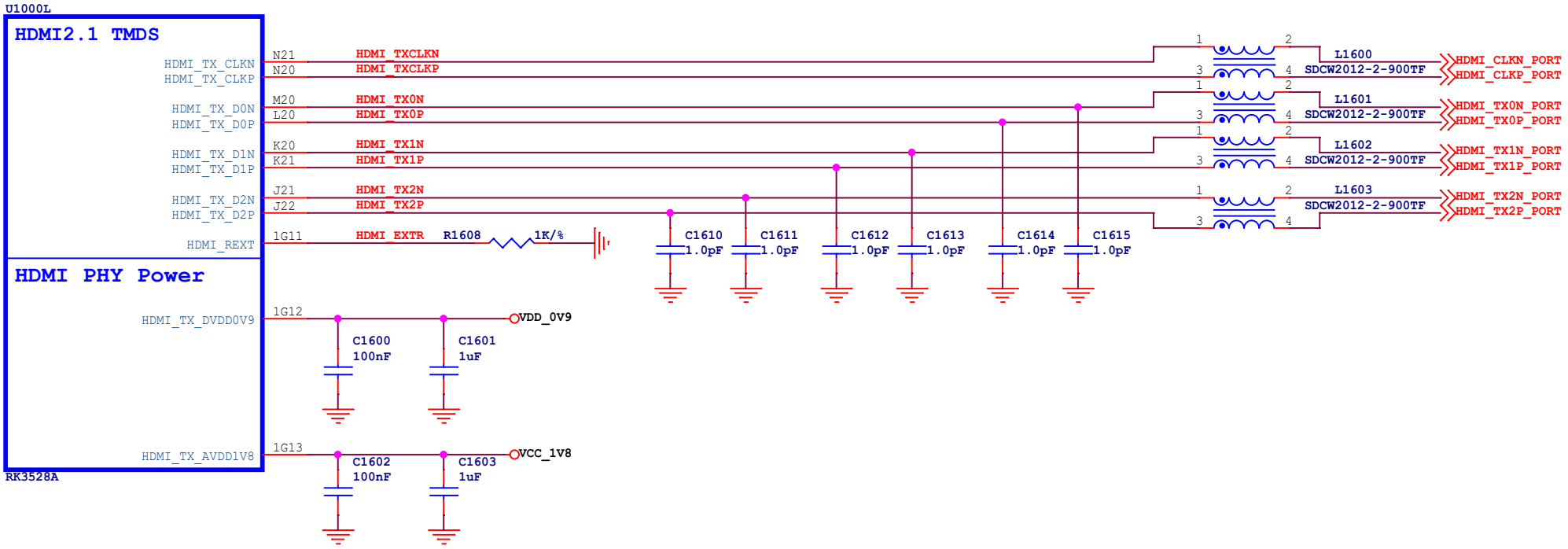

| HW_ID | Rup | Rdown | Note |
|---|---|---|---|
| HW_ID1 | 100K | 12K | onboard antenna |
| HW_ID2 | 100K | 27K | external antenna |




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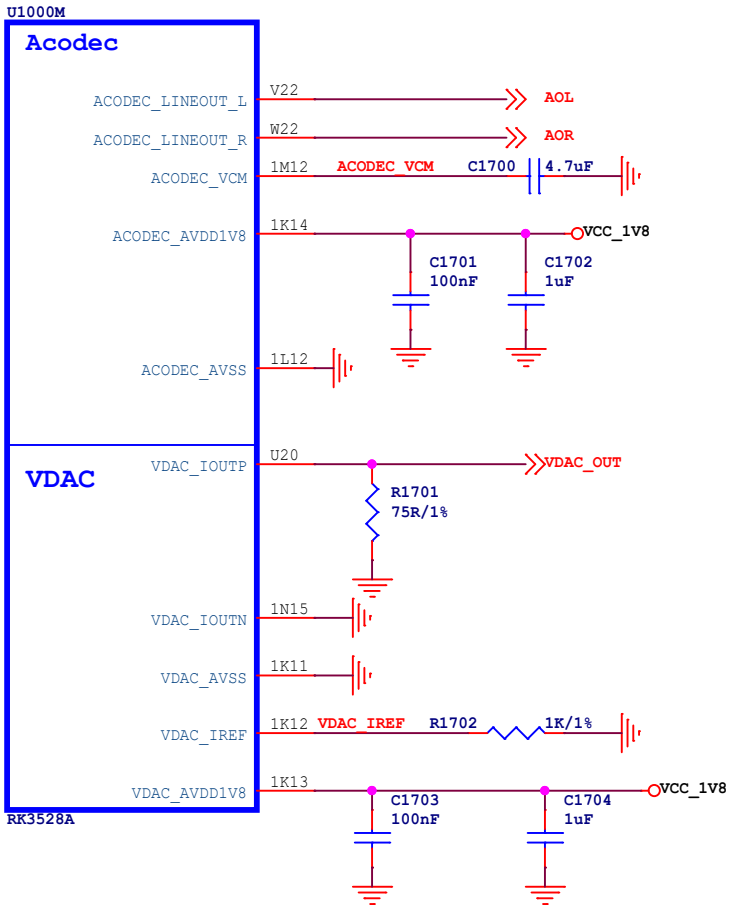
| Size | Title: | REV |
|---|---|---|
| A4 | ROCK 2A | V1.2 |
| | Page Name: 09.RK3528A SARADC/OTP | |
| Date: Thursday, June 06, 2024 | Sheet 09 of 25 | |

# RK3528A_L(HDMI PHY)



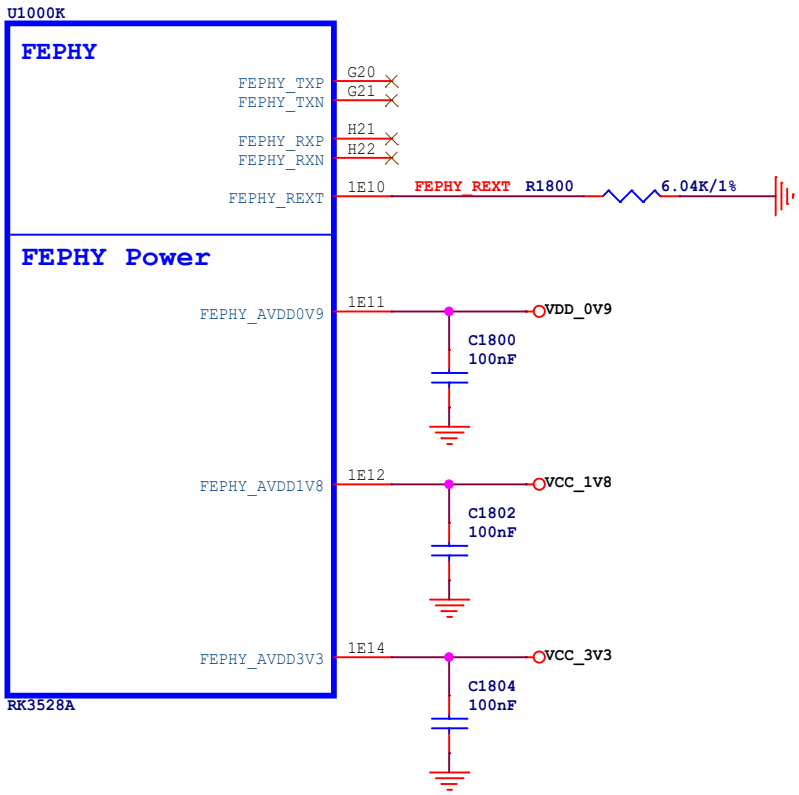

| | | |
|---|---|---|
| Size: A4 | Title: ROCK 2A | REV: V1.2 |
| | Page Name: 10.RK3528A HDMI Interface | |
| Date: Thursday, June 06, 2024 | Sheet 10 of 25 | |

# RK3528A_M(Acodec/VDAC)



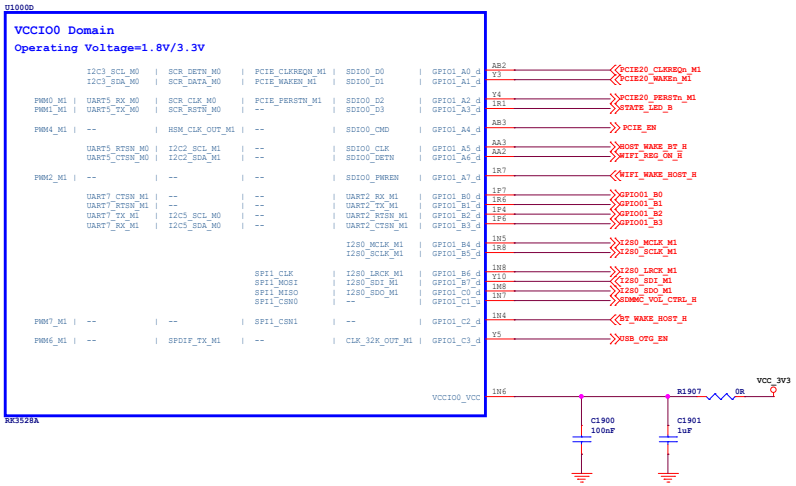

| | | |
|---|---|---|
| Size: A4 | Title: ROCK 2A | REV: V1.2 |
| | Page Name: 11.RK3528A AV OUT | |
| Date: Thursday, June 06, 2024 | Sheet 11 of 25 | |

# RK3528A_K(Embed FEPHY)

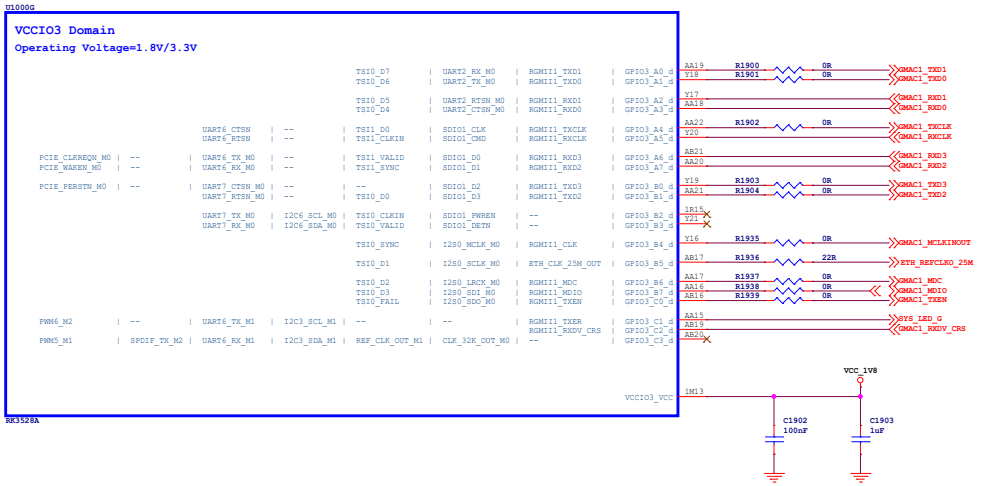



| | | | |
|---|---|---|---|
| Size | Title: | ROCK 2A | REV |
| A4 | Page Name: | 12.RK3528A Embed FEPHY | V1.2 |
| Date: | Thursday, June 06, 2024 | Sheet 12 of 25 | |

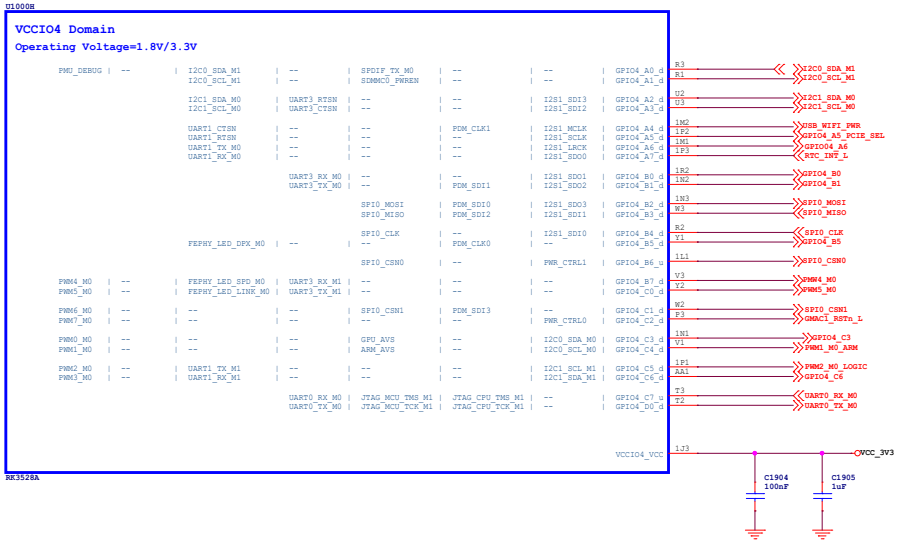
# RK3528A_D(VCCIO0 Domain)

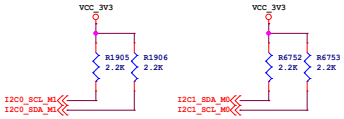



# RK3528A_G(VCCIO3 Domain)



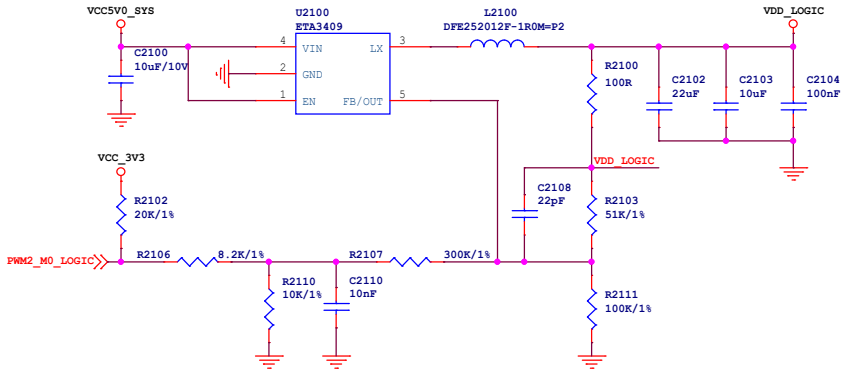

# RK3528A_H(VCCIO4 Domain)

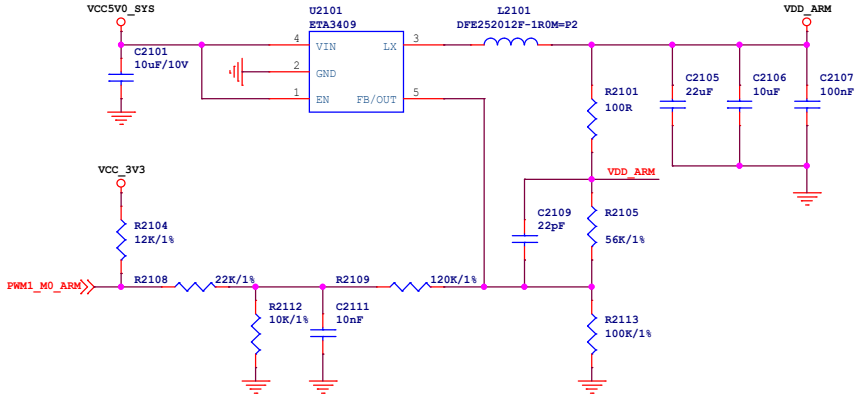


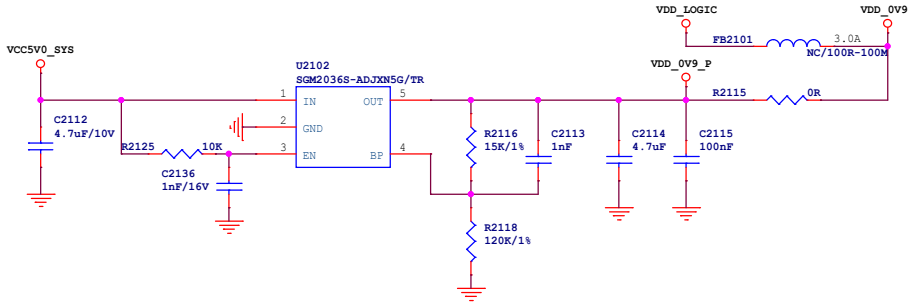



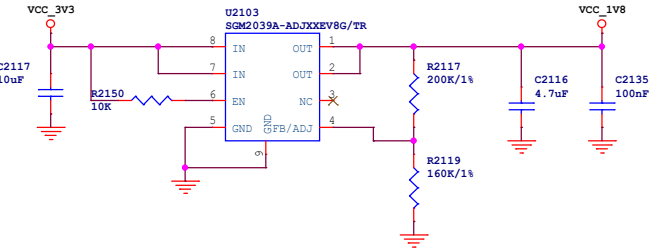

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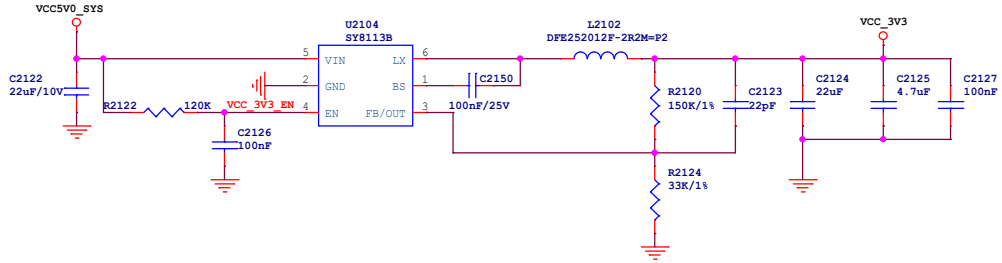
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|---|---|---|---|
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| Date: | Thursday, June 06, 2024 | Sheet 13 of 25 | |

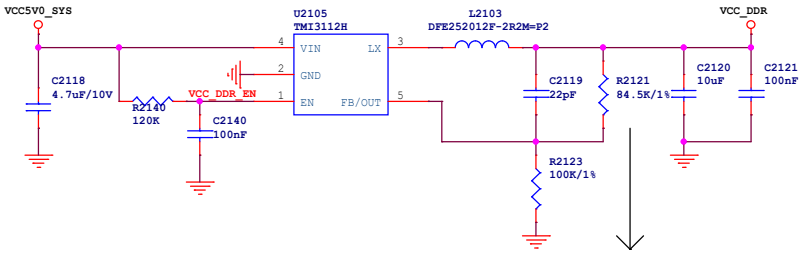


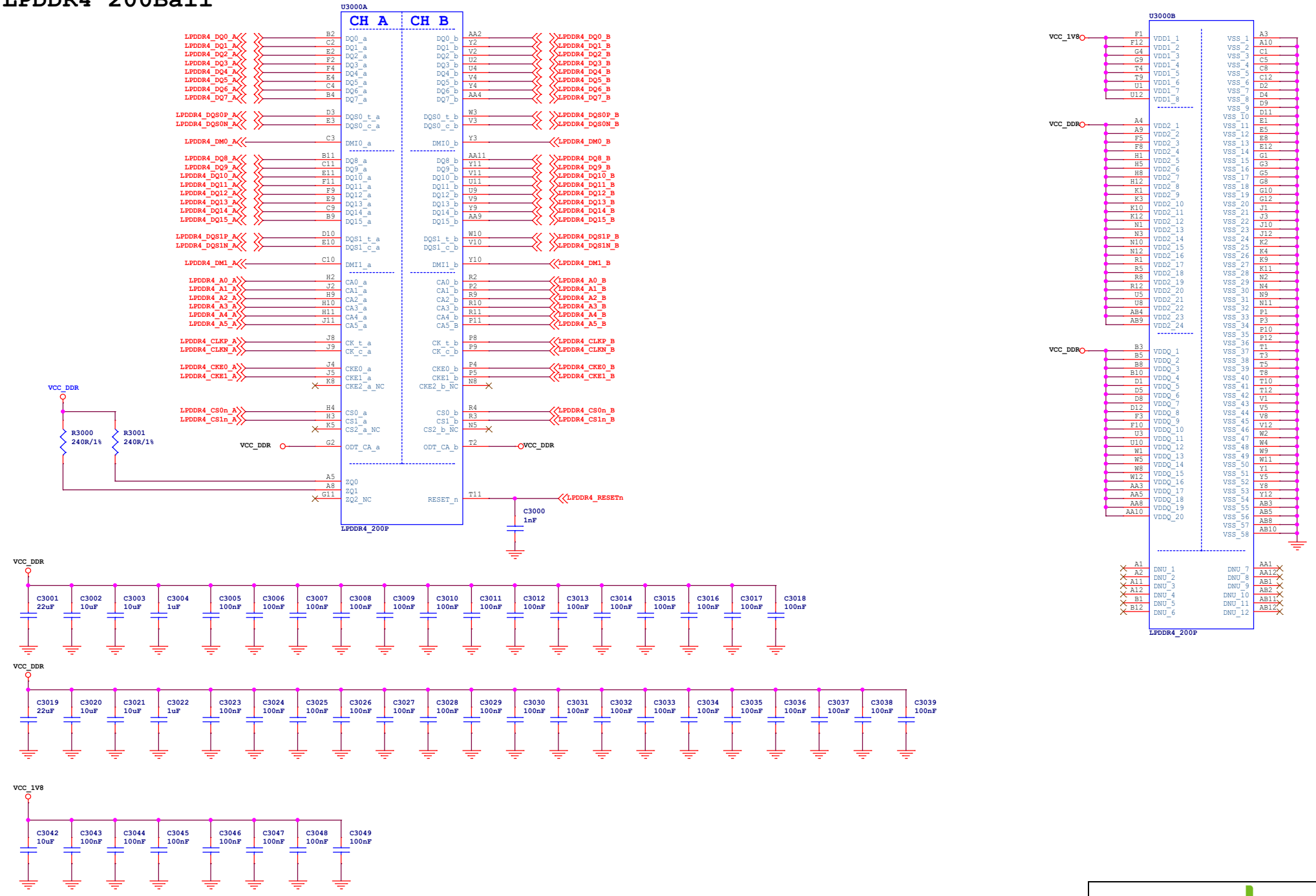




| | | | |
|---|---|---|---|
| DDR4 | 1.20V | 100K | 1% |
| LPDDR4/4x | 1.10V | 84.5K | 1% |
| DDR3 | 1.5V | 150K | 1% |
| DDR3L | 1.35V | 127K | 1% |
| LPDDR3 | 1.25V | 110K | 1% |

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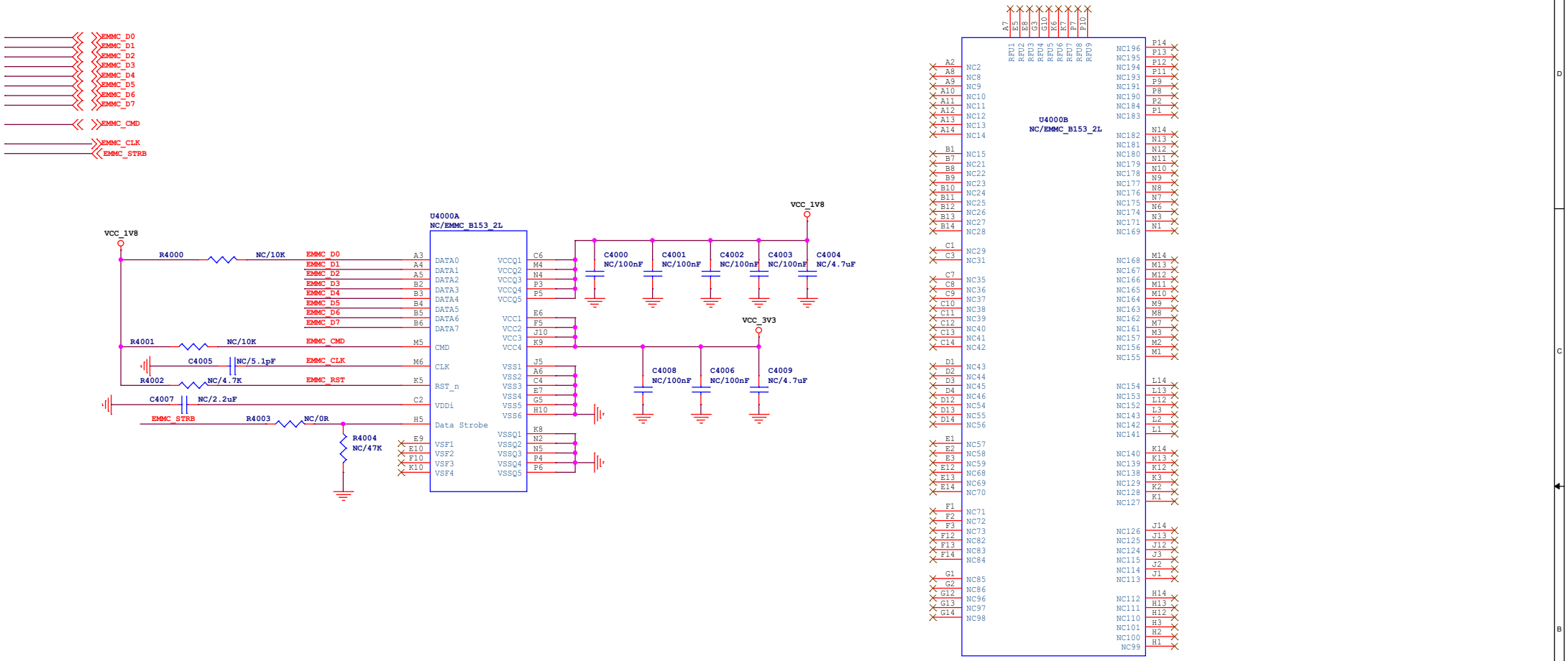
| | | |
|---|---|---|
| Size | Title: ROCK 2A | REV |
| A3 | Page Name: 14.Power-DiscretePower | V1.2 |
| Date: Thursday, June 06, 2024 | Sheet 14 of 25 | |

# LPDDR4 200Ball



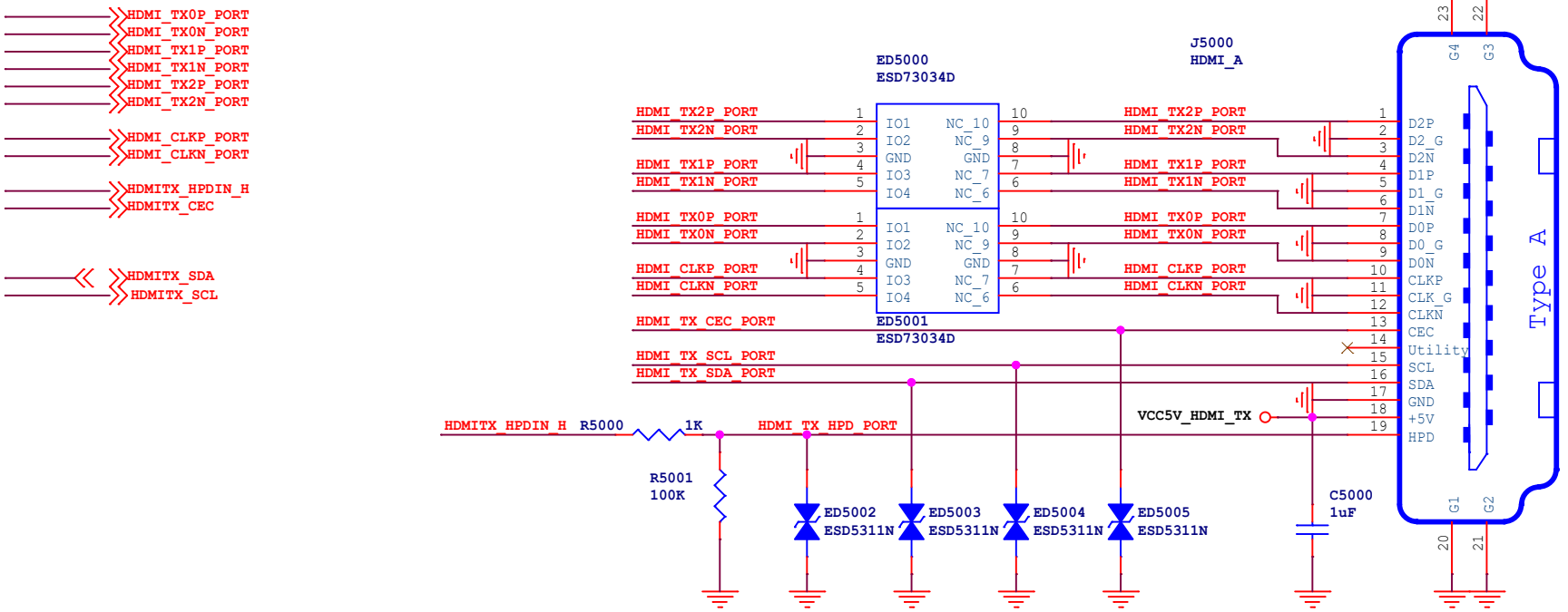

| | | | |
|---|---|---|---|
| Size | Title: ROCK 2A | | REV |
| A3 | Page Name: 15.DRAM-LPDDR4_1X32bit_200P | | V1.2 |
| Date: Thursday, June 06, 2024 | Sheet 15 of 25 | | |

# EMMC (NC)

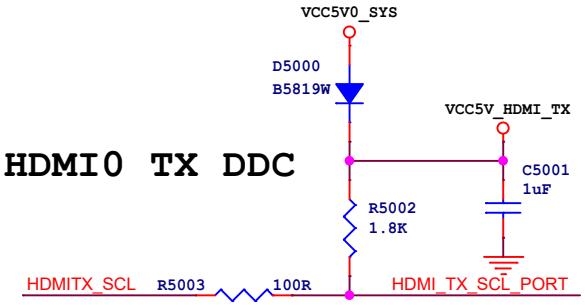


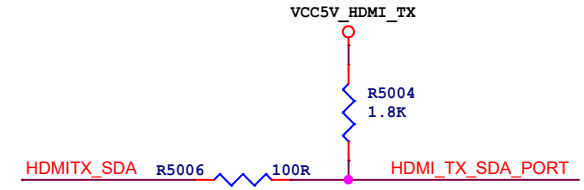

| | | | |
|---|---|---|---|
| Size | Title: ROCK 2A | | REV |
| A3 | Page Name: 16.Flash-eMMC | | V1.2 |
| Date: Thursday, June 06, 2024 | Sheet 16 of 25 | | |

# HDMI2.0 TX




## HDMI0 TX DDC



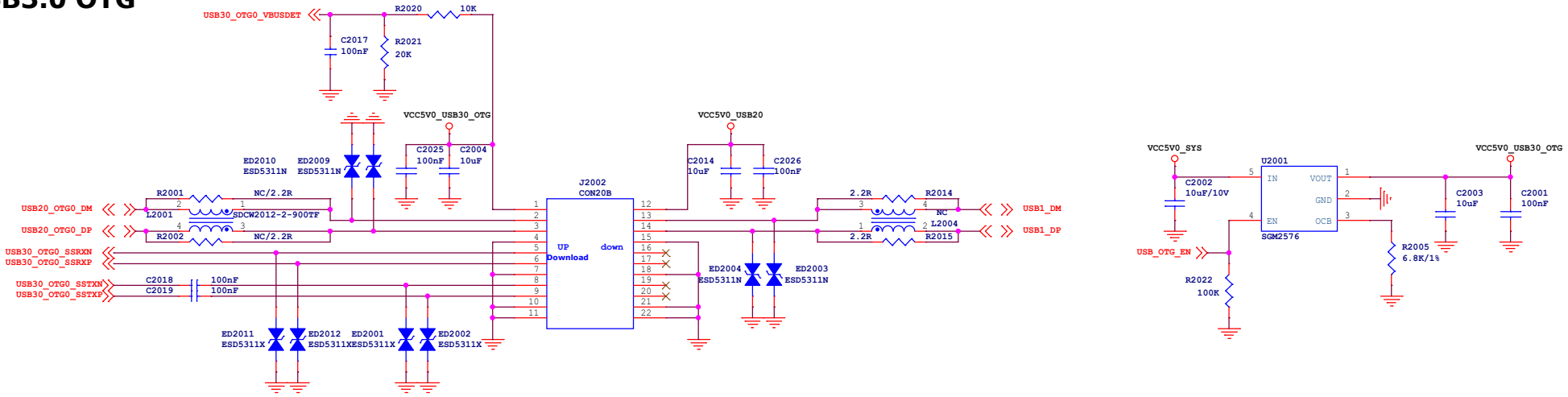




## HDMI0 TX CEC

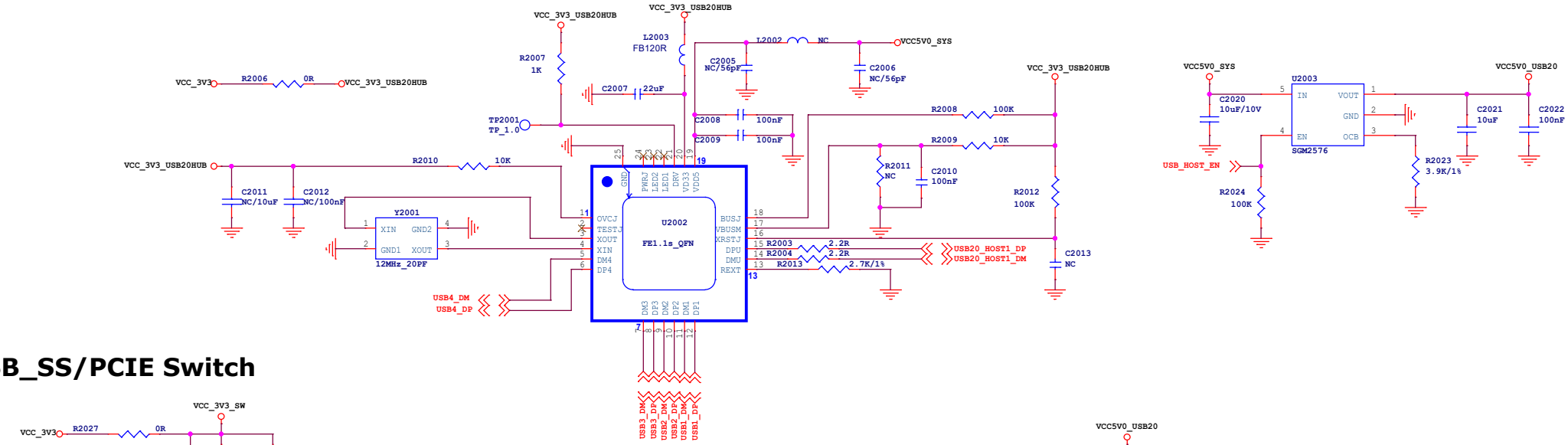



| | | | |
|---|---|---|---|
| Size | Title: ROCK 2A | | REV |
| A4 | Page Name: 17.HDMI TX | | V1.2 |
| Date: Thursday, June 06, 2024 | | Sheet 17 of 25 | |

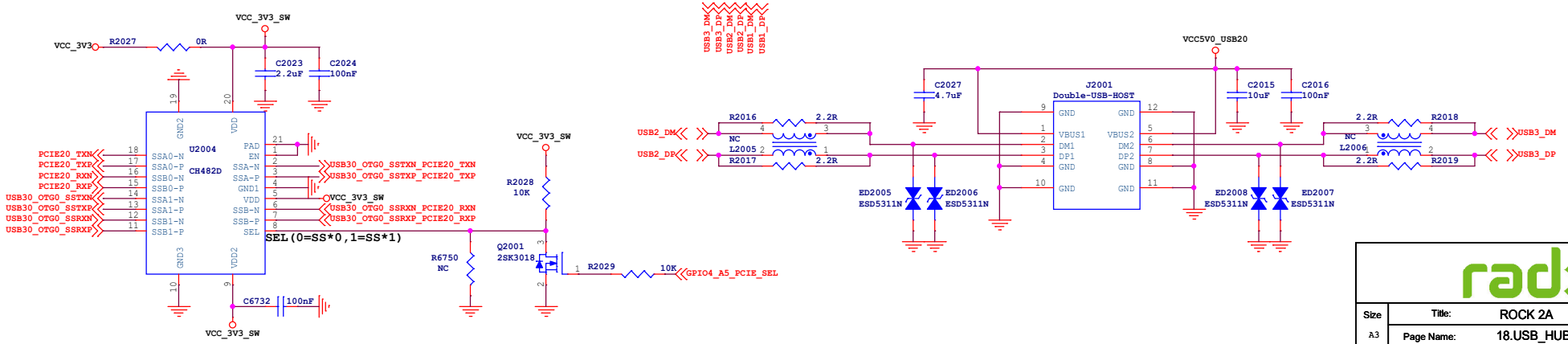
# USB3.0 OTG




# USB2.0 HUB+USB2.0




# USB_SS/PCIE Switch




| | | |
|---|---|---|
| Size A3 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 18.USB_HUB | |
| Date: Thursday, June 06, 2024 | Sheet 18 of 25 | |

# RGMII

GMAC1_TXD0
GMAC1_TXD1
GMAC1_TXD2
GMAC1_TXD3
GMAC1_TXEN
GMAC1_TXCLK
GMAC1_RXD0
GMAC1_RXD1
GMAC1_RXD2
GMAC1_RXD3
GMAC1_RXDV_CRS
GMAC1_RXCLK
ETH_REFCLKO_25M
GMAC1_MCLKINOUT
GMAC1_MDC
GMAC1_MDIO
GMAC1_RSTn_L

Y6700
PHY_XTALOUT
C6700 18pF
XIN GND2 GND1 XOUT
25MHz
C6701 18pF
PHY_XTALIN

ETH_REFCLKO_25M R6708 NC/0R PHY_XTALOUT

GMAC1_MCLKINOUT R6710 NC/100R PHY_CLKOUT125
C6703 DNP
R6712 NC/120R

# RJ45

MDI0+ MDI0- FB6701 BCMEF122P900H PHY1_MDI0+ PHY1_MDI0-
MDI1+ MDI1- FB6702 BCMEF122P900H PHY1_MDI1+ PHY1_MDI1-
MDI2+ MDI2- FB6703 BCMEF122P900H PHY1_MDI2+ PHY1_MDI2-
MDI3+ MDI3- FB6704 BCMEF122P900H PHY1_MDI3+ PHY1_MDI3-

J6703
HR911130C
PHY1_MDI0+ PHY1_MDI0- PHY1_MDI1+ PHY1_MDI1- PHY1_MDI2+ PHY1_MDI2- PHY1_MDI3+ PHY1_MDI3-
MDI0+ MDI0- MDI1+ MDI1- MDI2+ MDI2- MDI3+ MDI3-
TCT TCT1 G+ G- Y+ Y-
VC1 VC2 VC3 VC4

C6726 100nF/50V
PHY_LED1/CFG_LDO0
VCC3V3_PHY
R6751 510R
R6713 510R
R6714 NC
R6715 0R
R6716 NC
PHY_LED2/CFG_LDO1

J6701 CON4A
TR0 TR1 TR2 TR3

VDD10_PHY
R6722 2.49K
VCC3V3_PHY
R6749 4.7K VCC3V3_PHY
RSETO PHY_XTALOUT PHY_XTALIN PHY_CLKOUT125 PHY_LED2/CFG_LDO1 PHY_LED1/CFG_LDO0 PHY_LED0/CFG_EXT

U6700
RTL8211F-CG
E_Pad AVDD33 RSETO AVDD10 XTAL_IN/EXT_CLKIN XTAL_OUT CLKOUT LED2/CFG_LDO1 LED1/CFG_LDO0 LED0/CFG_EXT INTB/PMEB
MDI0+ MDI0- MDI1+ MDI1- MDI2+ MDI2- MDI3+ MDI3-
MDI[0]+ MDI[0]- AVDD10 MDI[1]+ MDI[1]- MDI[2]+ MDI[2]- AVDD10 MDI[3]+ MDI[3]-
REG_OUT DVDD33 DVDD_RG RXC/PHYAD1 RXCTL/PHYAD2 RXD0/RXDLY RXD1/TXDLY RXD2/PLLOFF RXD3/PHYAD0 DVDD10
PHY_REG_OUT VCC3V3_PHY VCCIO_PHY PHY_RXCLK/PHYAD1 PHY_RXDV/PHYAD2 PHY_RXD0/RXDLY PHY_RXD1/TXDLY PHY_RXD2/PLLOFF PHY_RXD3/PHYAD0 VDD10_PHY
AVDD33 PHYRSTB MDC MDIO TXD3 TXD2 TXD1 TXD0 TXCTL TXC
PHY_RSTn GMAC1_MDC GMAC1_MDIO GMAC1_TXD3 GMAC1_TXD2 GMAC1_TXD1 GMAC1_TXD0 GMAC1_TXEN GMAC1_TXCLK

VCC3V3_PHY
C6710 NC/3pF
R6736 NC/4.7K
R6737 1.8K VCCIO_PHY
C6717 100nF
GMAC1_RSTn_L R6738 0R

PHY_RXD0/RXDLY R6739 22R GMAC1_RXD0
PHY_RXD1/TXDLY R6740 22R GMAC1_RXD1
PHY_RXD2/PLLOFF R6741 22R GMAC1_RXD2
PHY_RXD3/PHYAD0 R6742 22R GMAC1_RXD3
PHY_RXCLK/PHYAD1 R6743 22R GMAC1_RXCLK
PHY_RXDV/PHYAD2 R6744 22R GMAC1_RXDV_CRS
C6723 DNP

R6717 4.7K PHY_LED0/CFG_EXT R6718 NC/4.7K VCC3V3_PHY
R6719 NC/4.7K PHY_LED2/CFG_LDO1 R6720 4.7K
R6721 4.7K PHY_LED1/CFG_LDO0

R6723 NC/4.7K PHY_RXD3/PHYAD0 R6724 4.7K VCCIO_PHY
R6725 4.7K PHY_RXCLK/PHYAD1
R6726 4.7K PHY_RXDV/PHYAD2

R6727 NC/4.7K PHY_RXD0/RXDLY R6728 4.7K VCCIO_PHY
R6729 4.7K PHY_RXD1/TXDLY R6730 NC/4.7K VCCIO_PHY
R6732 NC/4.7K PHY_RXD2/PLLOFF R6733 4.7K VCCIO_PHY

L6700 DFE252012F-2R2M=P2
PHY_REG_OUT
VDD10_PHY
C6714 4.7uF
C6715 100nF
C6716 100nF
C6720 100nF
C6724 100nF
C6725 100nF

PHY_LED2/CFG_LDO1 C6705 NC/100pF
PHY_LED1/CFG_LDO0 C6706 NC/100pF

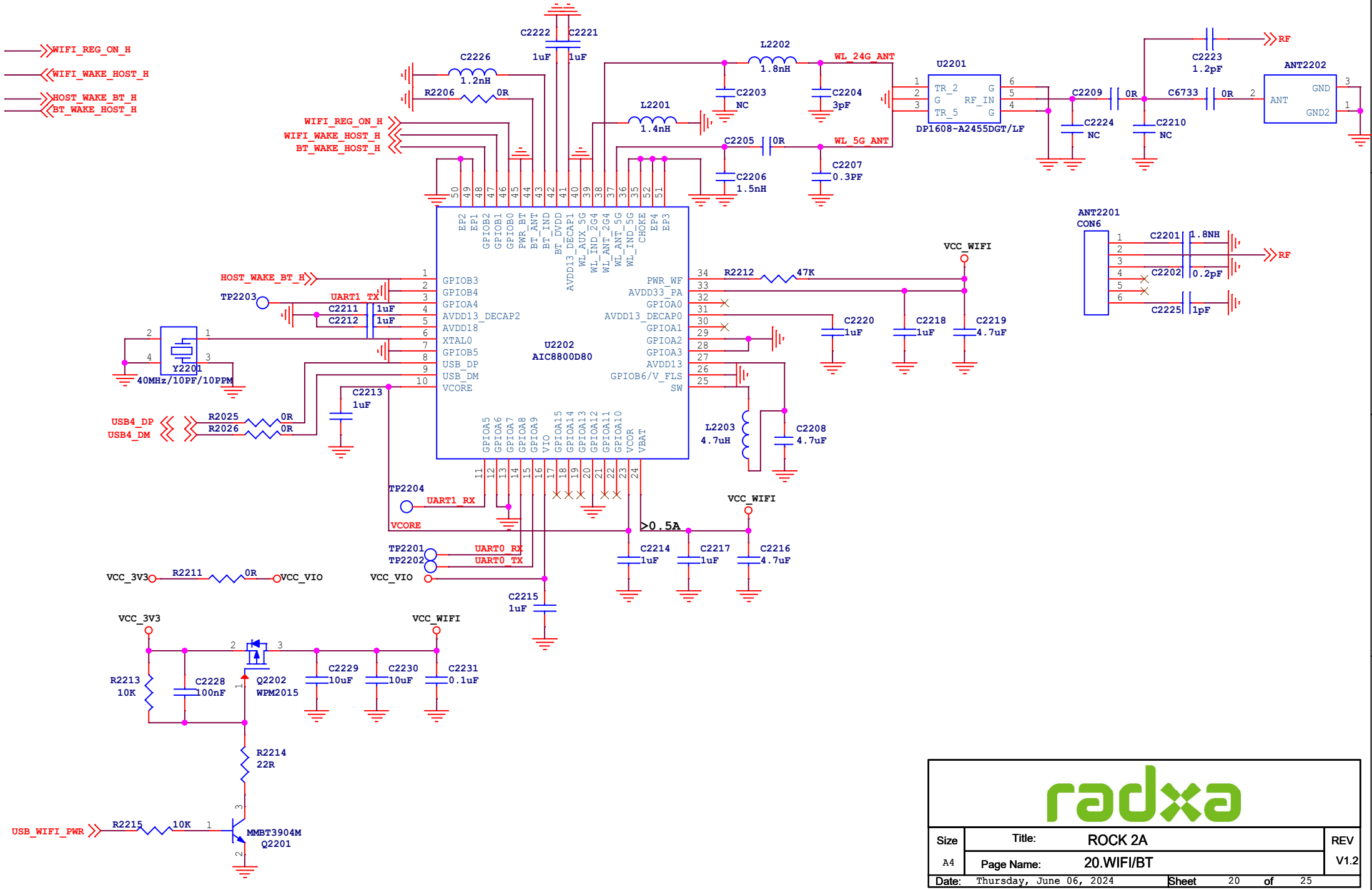
VCC_1V8 R6731 NC/0.1R
VCC_3V3 R6734 NC/0.1R
VCCIO_PHY
C6707 4.7uF
C6708 1uF
C6709 100nF

VCC_3V3 R6735 0.1R
VCC3V3_PHY
C6711 4.7uF
C6712 100nF
C6713 100nF
C6718 4.7uF
C6719 100nF

radxa

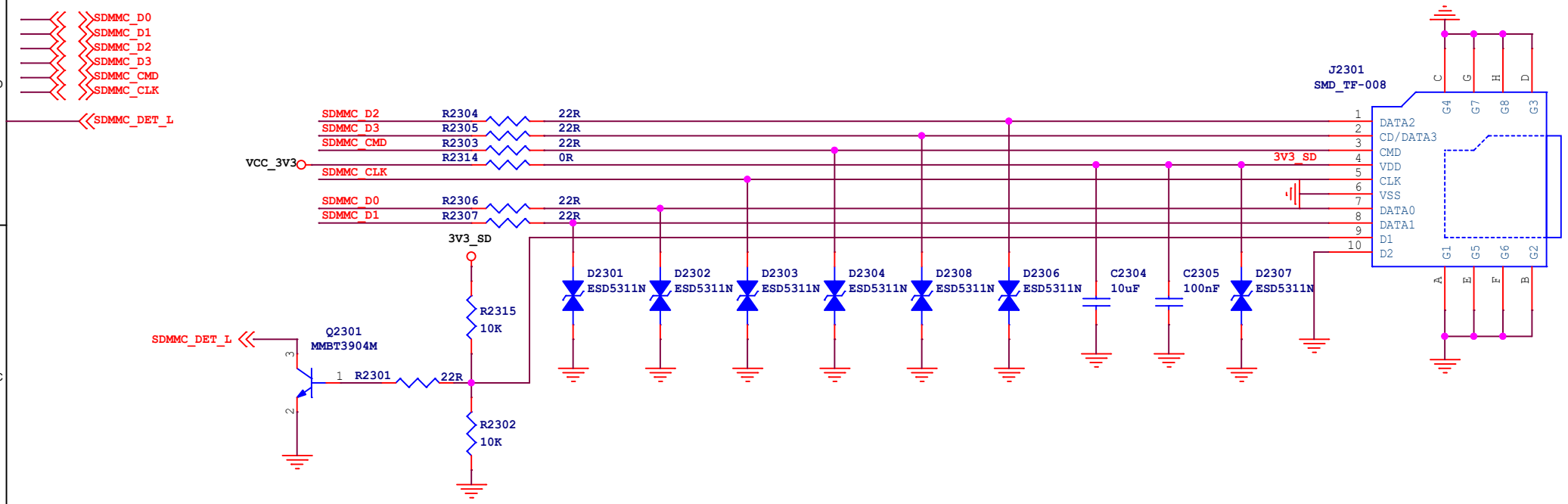
| Size | Title: | REV |
|---|---|---|
| A3 | ROCK 2A | V1.2 |
| | Page Name: 19.Ethernet_RGMII | |
| Date: Thursday, June 06, 2024 | Sheet 19 of 25 | |

# WIFI/BT(USB interface)

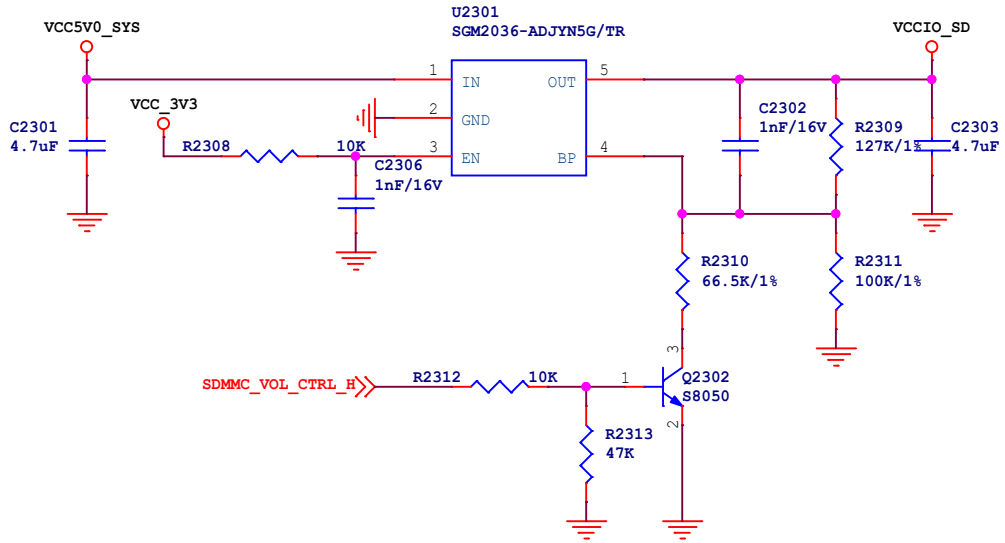



| | | |
|---|---|---|
| Size A4 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 20.WIFI/BT | |
| Date: Thursday, June 06, 2024 | Sheet 20 of 25 | |

# Micro_SD Card

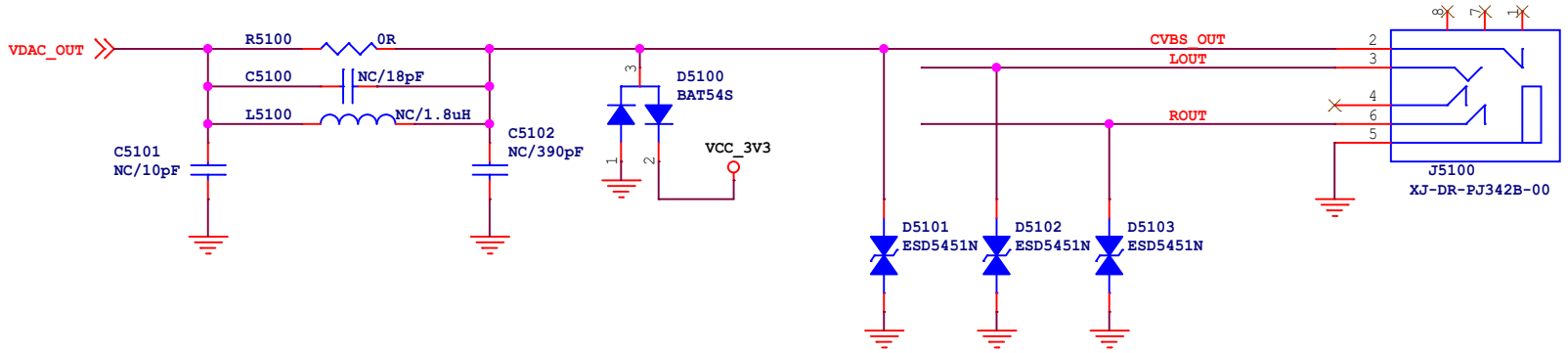



# VCCIO_SD

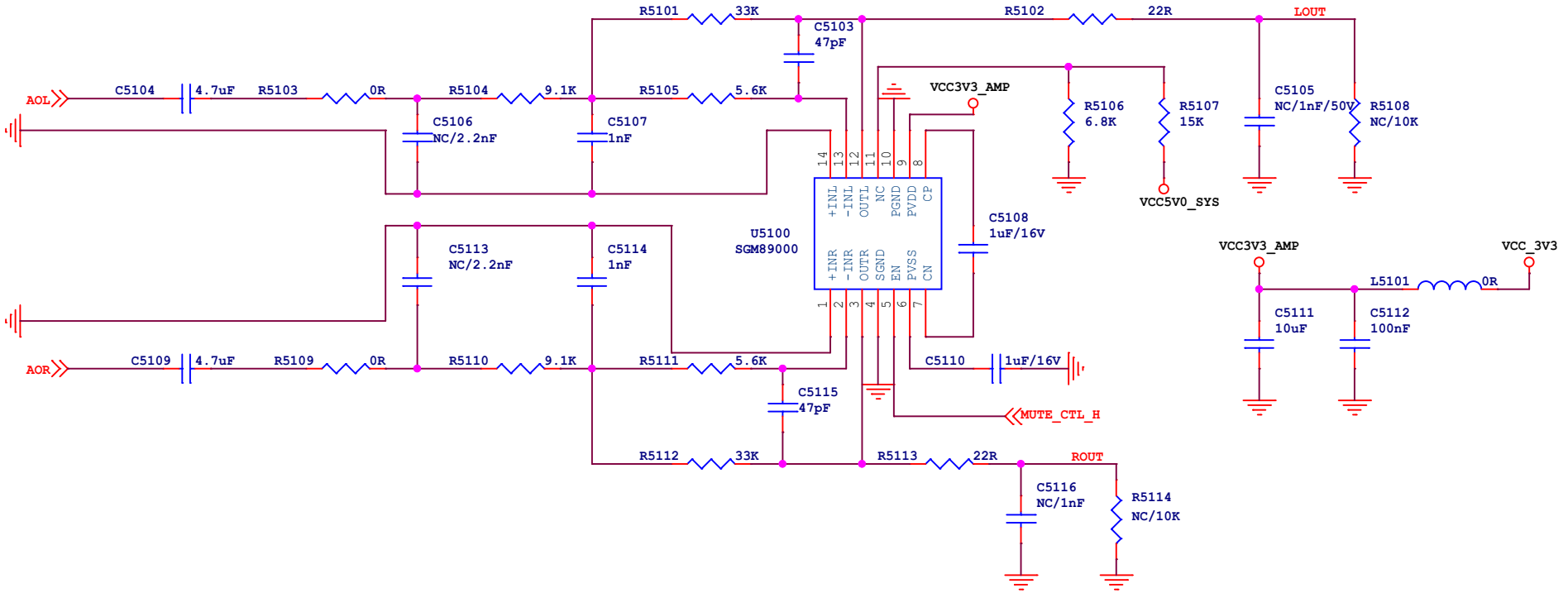



| | | | |
|---|---|---|---|
| Size A4 | Title: ROCK 2A | | REV V1.2 |
| | Page Name: 21.Micro_SD | | |
| Date: Thursday, June 06, 2024 | | Sheet 21 of 25 | |

# CVBS/AV_OUT

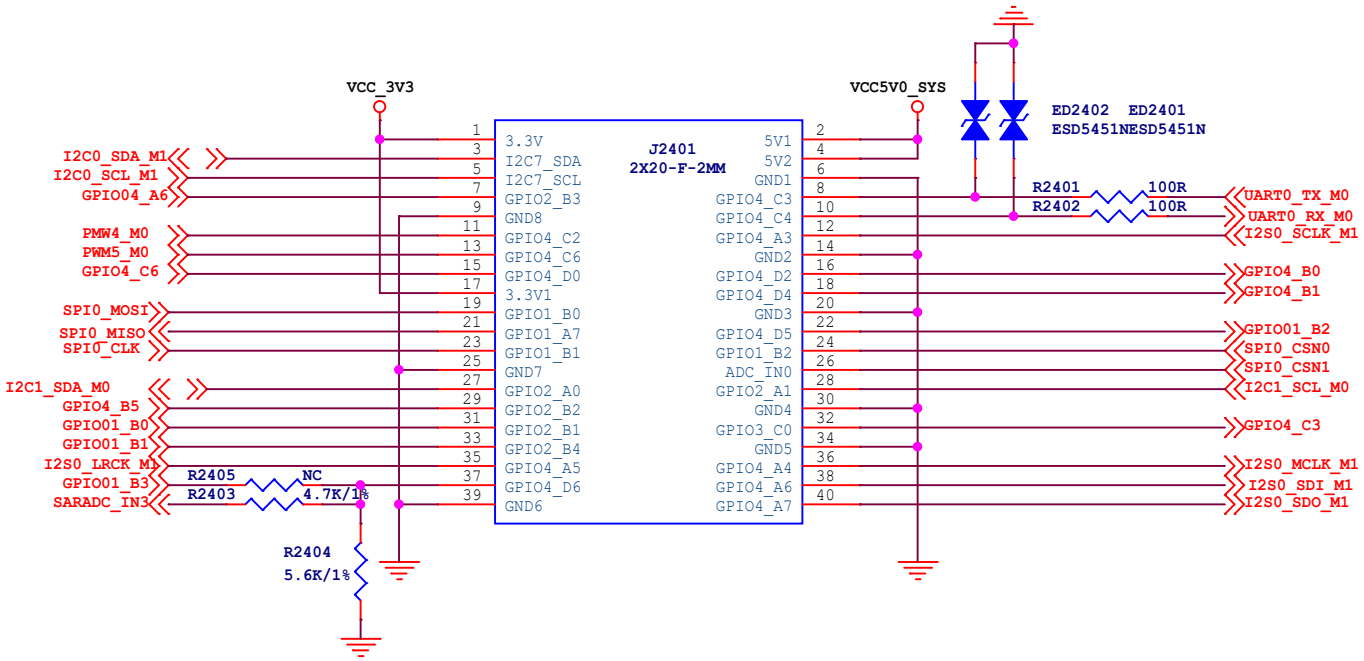


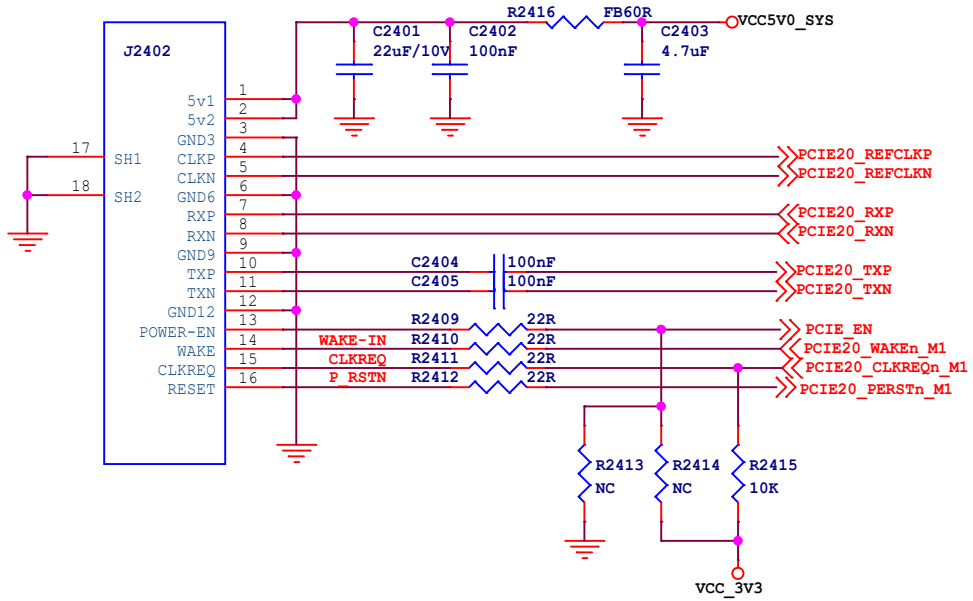

# 2-Vrms Audio Line Driver



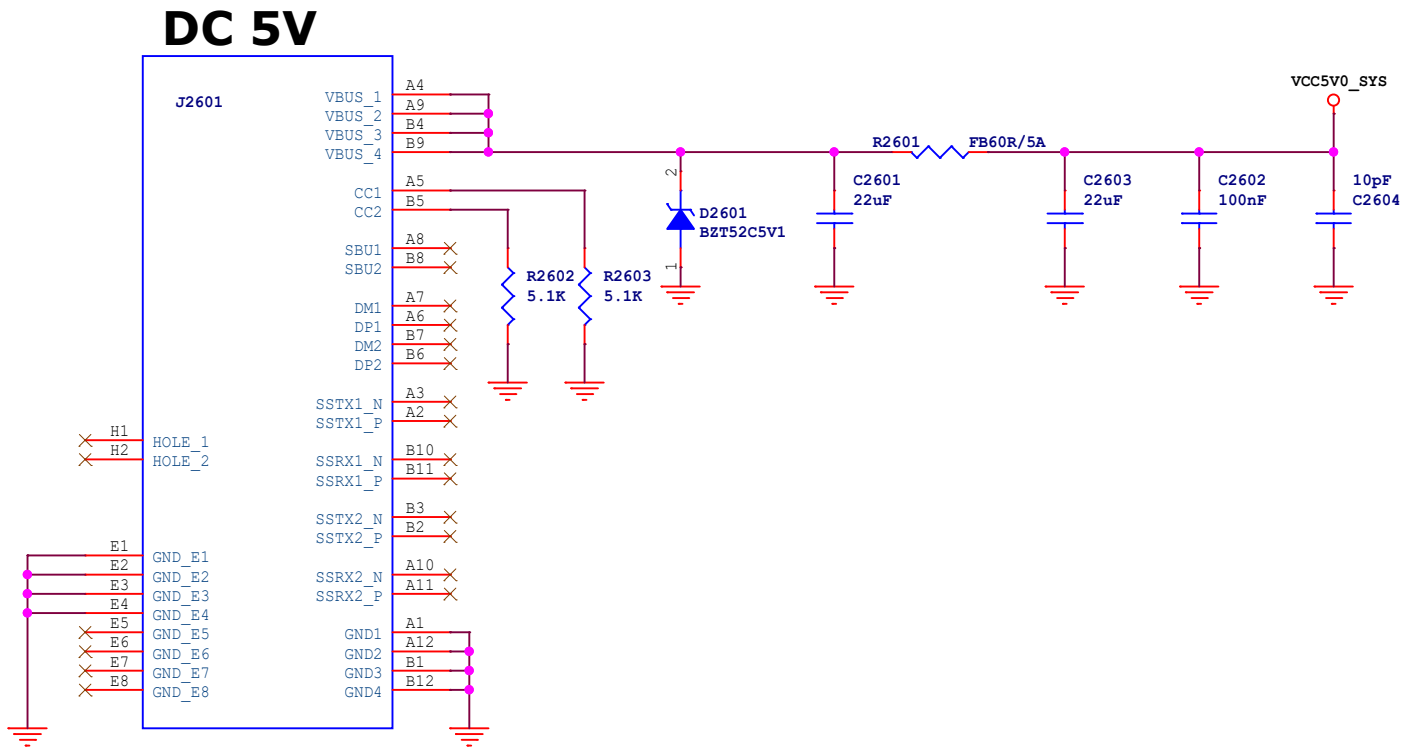

| | | | |
|---|---|---|---|
| Size | Title: | ROCK 2A | REV |
| A4 | Page Name: | 22.VO-AV Interface | V1.2 |
| Date: | Thursday, June 06, 2024 | Sheet 22 of 25 | |

# 40PIN_Connctor/PCIE interface



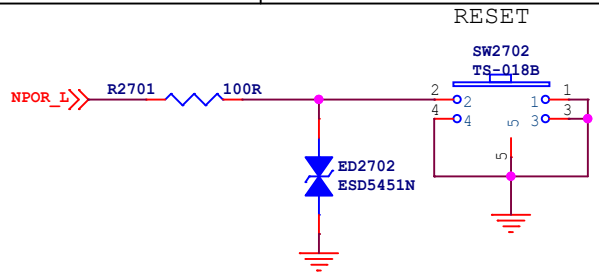




| | | |
|---|---|---|
| Size A4 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 23.40PIN_Connctor/PCIE interface | |
| Date: Thursday, June 06, 2024 | Sheet 23 of 25 | |

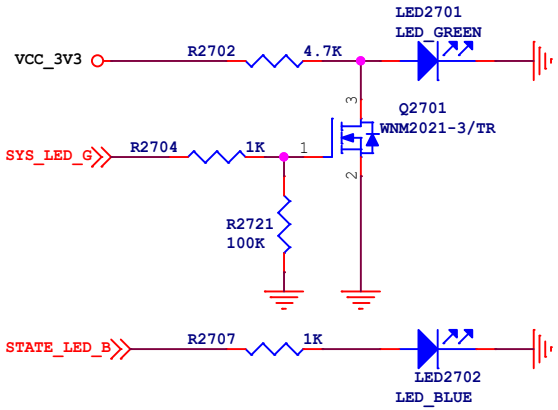



| | | |
|---|---|---|
| Size: A4 | Title: ROCK 2A | REV: V1.2 |
| | Page Name: 24.5V_Power | |
| Date: Thursday, June 06, 2024 | Sheet 24 of 25 | |

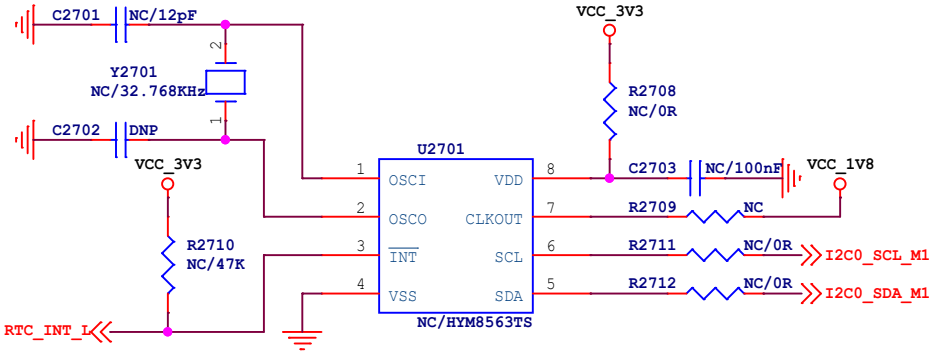
# KEY



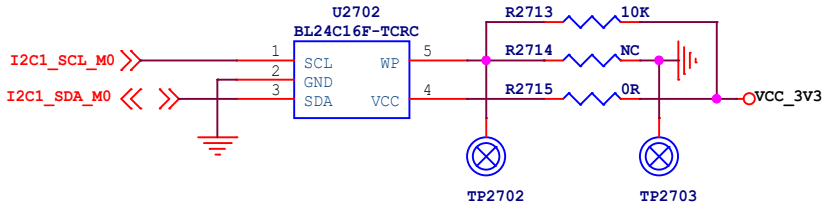

# LED



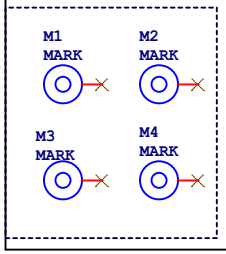

# RTC(option)




# EEPROM




# MARK




| | | |
|---|---|---|
| Size A4 | Title: ROCK 2A | REV V1.2 |
| | Page Name: 25.Key_LED_RTC_EEPROM | |
| Date: Thursday, June 06, 2024 | Sheet 25 of 25 | |